

Chapter #4: Memory System

- Memory is an essential component of the microcomputer system.
- It stores binary instructions and data for the microcomputer.
- The memory is the place where the computer holds current programs and data that are in use.
- Computer memory exhibits perhaps the widest range of type, technology, organization, performance and cost of any feature of a computer system.
- The memory unit that communicates directly with the CPU is called main memory.
- Devices that provide backup storage are called auxiliary memory or secondary memory.

Memory System Characteristics

1. Location:

Processor memory: The memory like registers is included within the processor and termed as processor memory.

Internal memory: It is often termed as main memory and resides within the CPU.

External memory: It consists of peripheral storage devices such as disk and magnetic tape that are accessible to processor via i/o controllers.

2. Capacity:

Word size: Capacity is expressed in terms of words or bytes.

— The natural unit of organization

Number of words: Common word lengths are 8, 16, 32 bits etc. — or Bytes

3. Unit of Transfer:

Internal: For internal memory, the unit of transfer is equal to the number of data lines into and out of the memory module.

External: For external memory, they are transferred in block which is larger than a word.

4. Access Method:

Sequential access: In this access, it must start with beginning and read through a specific linear sequence. Example: Tape

Direct Access: Individual blocks of records have unique address based on location.

Example: Disk

Random access: The time to access a given location is independent of the sequence of prior accesses and is constant. Example: RAM

Associative access: This is random access type of memory that enables one to make a comparison of desired bit locations within a word for a specified match, and to do this for all words simultaneously. Example: Cache

5. Performance:

----Access time:

For random access memory, access time is the time it takes to perform a read or write operation

Whereas for non-random access, it is the time needed to position read / write mechanism at desired location.

– Memory Cycle Time:

It is the total time that is required to store next memory access operation from the previous memory access operation.

Memory cycle time = access time plus transient time (any additional time required before a second access can commence).

– **Transfer Rate:**

This is the rate at which data can be transferred in and out of a memory unit.

Rate at which data can be moved

For random access, $R = 1 / \text{cycle time}$

For non-random access, $T_n = T_a + N / R$

- where T_n – average time to read or write N bits, T_a – average access time, N – number of bits, R – Transfer rate in bits per second (bps).

- **Physical Type:**

Semiconductor - RAM

Magnetic – Disk or Tape

Optical – CD or DVD

Others – Bubble (thin film) or Hologram (laser)

- **Physical Characteristics:**

Decay: Information decays mean data loss.

Volatility: Information decays when electrical power is switched off.

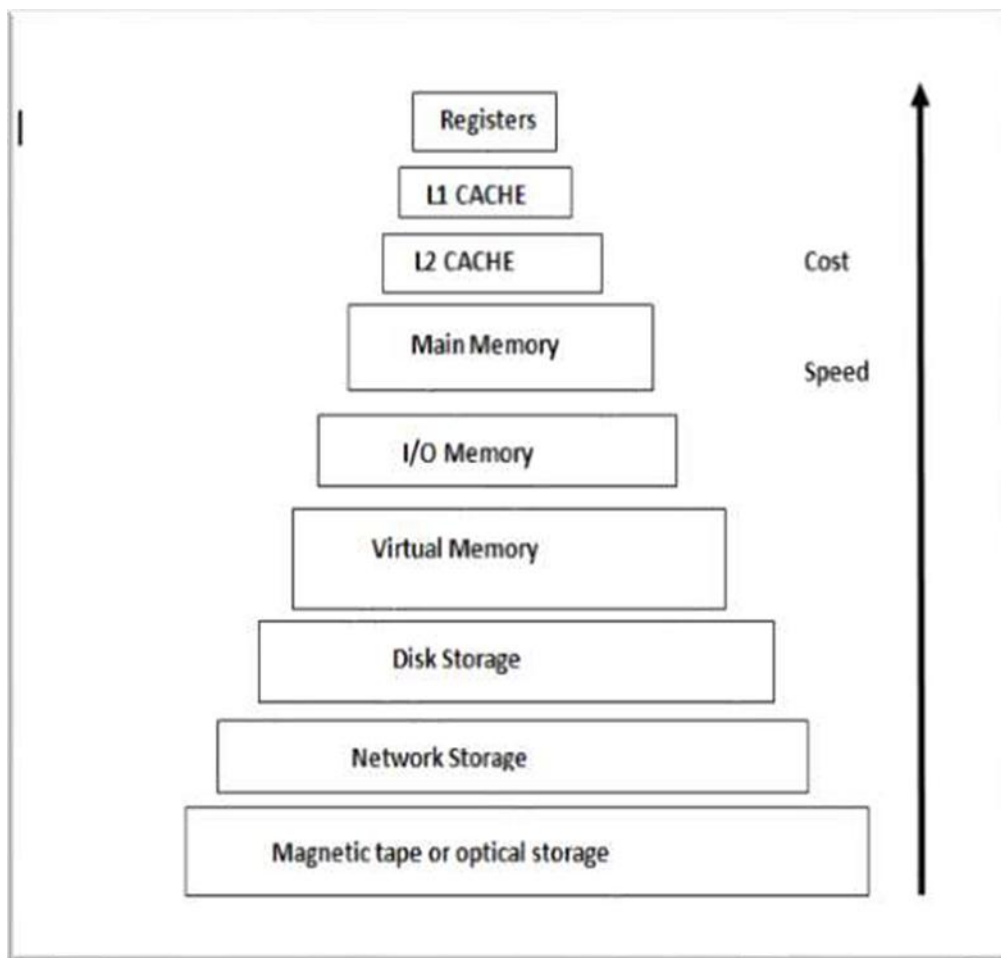
Erasable: Erasable means permission to erase.

Power consumption: how much power consumes?

Memory Hierarchy and Classification:

In the Computer System Design, Memory Hierarchy is an enhancement to organize the memory such that it can minimize the access time. The Memory Hierarchy was developed based on a program behavior known as locality of references (same data or nearby data is likely to be accessed again and again).

Memory Hierarchy helps in optimizing the memory available in the computer. There are multiple levels present in the memory, each one having a different size, different cost, etc.



- Capacity, Cost and Speed
- Tradeoff (One can not achieved)
 - If Capacity $\uparrow$ Access_Time $\uparrow$ Cost (per bit) $\downarrow$
 - If Access_Time $\downarrow$ Capacity $\downarrow$ Cost (per bit) $\uparrow$

Types of Computer Memory

In general, computer memory is divided into three types:

- Primary memory
- Secondary memory
- Cache memory

Now we discuss each type of memory one by one in detail:

1. Primary Memory

It is also known as the main memory of the computer system. It is used to store data and programs, or instructions during computer operations. It uses semiconductor technology and hence is commonly called semiconductor memory. Primary memory is of two types:

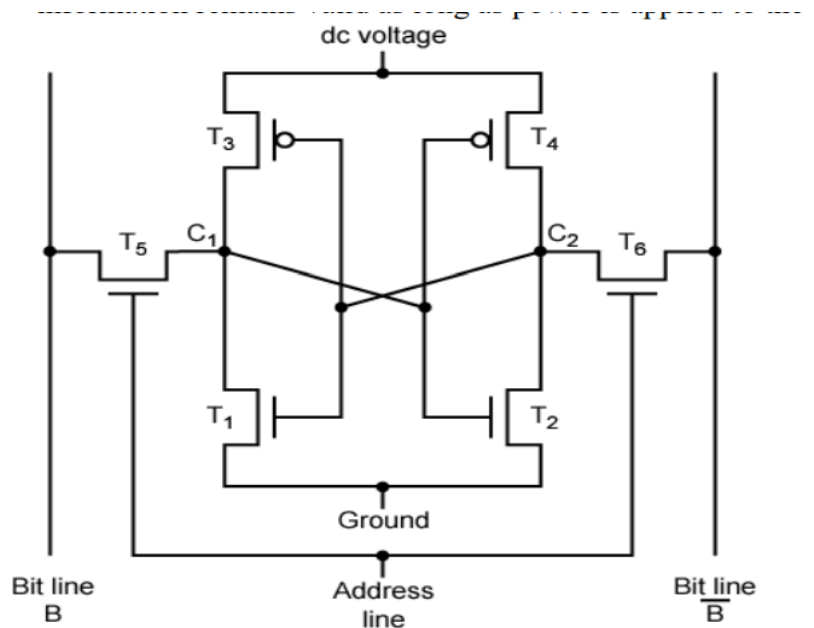
RAM (Random Access Memory):

It is a volatile memory. Volatile memory stores information based on the power supply. If the power supply fails/ interrupted/stopped, all the data and information on this memory will be lost. RAM is used for booting up or starting the computer. It temporarily stores programs/data which has to be executed by the processor. RAM is of two types:

- **S RAM (Static RAM):**

S RAM uses transistors and the circuits of this memory are capable of retaining their state as long as the power is applied. This memory consists of the number of flip flops with each flip flop storing 1 bit. It has less access time and hence, it is faster.

The SRAM is volatile which means the data stored in it gets all wiped out once the power supply is cut. SRAM comprised of flip flops. It consists of 4-6 transistors, once the flip flop stores the bit it keeps it stored until the opposite bit is stored in it.



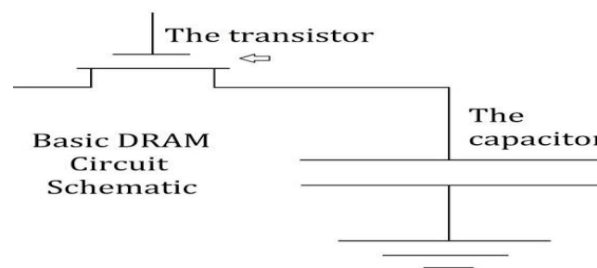
4 transistors → form **two cross-coupled inverters**

2 access (pass) transistors
→ connect the cell to bit lines

Working Of SRAM:

- Four transistors T1, T2, T3 and T4 are cross connected in an arrangement that produces a stable logical state. In logic state 1, point C1 is high and point C2 is low. In this state, T1 & T4 are off and T2 & T3 are on.
 - In logic state 0, point C1 is low and C2 is high. In this state, T1 & T4 are on and T2 & T3 are off.
 - The address line controls the two transistors T5 & T6. When a signal is applied to this line, the two transistors are switched on allowing for read and write operation.
 - For a write operation, the desired bit value is applied to line B while its complement is applied to line B complement.
 - This forces the four transistors T1, T2, T3 & T4 into a proper state.
 - For the read operation, the bit value is read from line B
- **D RAM (Dynamic RAM):** D RAM uses capacitors and transistors and stores the data as a charge on the capacitors. They contain thousands of memory cells. It needs refreshing of charge on capacitor after a few milliseconds.

- DRAM has to have its contents refreshed constantly to maintain data.
- It is cheaper compared to other forms of memory such as SRAM.
- Slower than SRAM.
- Power consumption in the case of DRAM is less than the other memory type but always greater in comparison to the case of SRAM.
- DRAM is physically smaller in comparison to other alternatives.
- It has short data lifetime.



ROM (Read Only Memory):

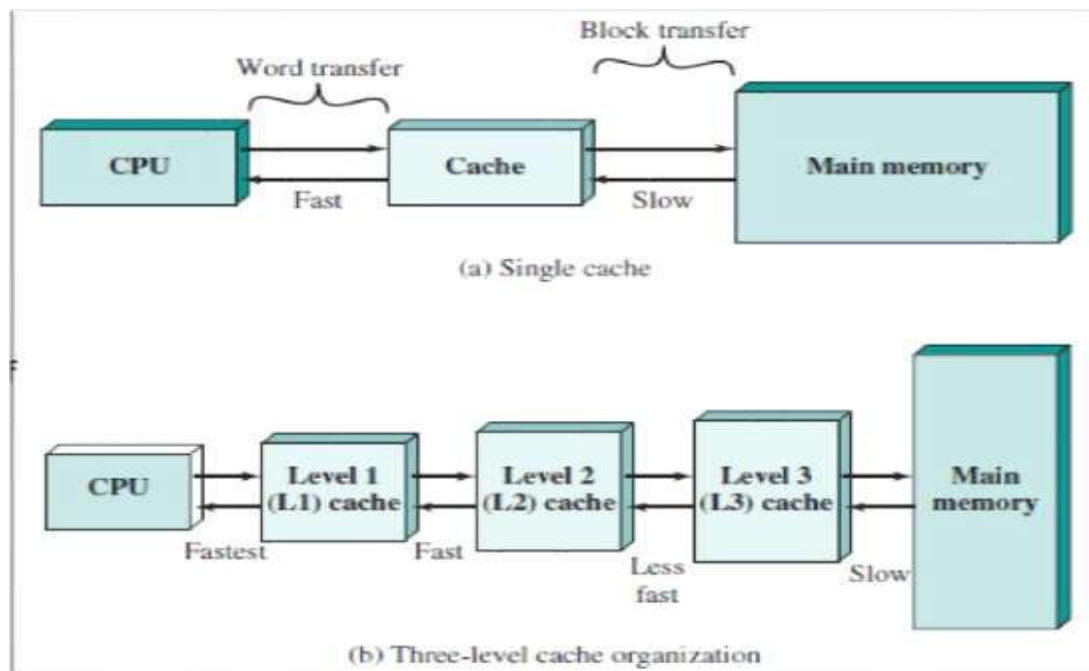
It is a non-volatile memory. Non-volatile memory stores information even when there is a power supply failed/ interrupted/stopped. ROM is used to store information that is used to operate the system. As its name refers to read-only memory, we can only read the programs and data that are stored on it. It contains some electronic fuses that can be programmed for a piece of specific information. The information is stored in the ROM in binary format. It is also known as permanent memory. ROM is of four types:

- **MROM(Masked ROM):** Hard-wired devices with a pre-programmed collection of data or instructions were the first ROMs. Masked ROMs are a type of low-cost ROM that works in this way.
- **PROM (Programmable Read Only Memory):**

- This read-only memory is modifiable once by the user. The user purchases a blank PROM and uses a PROM program to put the required contents into the PROM. Its content can't be erased once written.
 - PROM stands for Programmable Read-Only Memory. It is a memory chip on which information can be composed as it were once. Once a program has been composed onto a PROM, it remains there until the end of time.
- **EPROM (Erasable Programmable Read Only Memory):**
 - EPROM is an extension to PROM where you can erase the content of ROM by exposing it to Ultraviolet rays for nearly 40 minutes.
 - It is a memory chip that is non-volatile in nature that is it can hold the data even after the power supply is stopped.
 - It can be reused again and again as it is easily programmable and erasable. A chip that uses ultraviolet (UV) light to write and erase data is called an EPROM.
 - The chip is exposed to UV radiation when it becomes essential to erase the data, which resets all of the transistors to their initial states.
 - **EEPROM (Electrically Erasable Programmable Read Only Memory):**
 - EEPROM is a type of non-volatile primary memory and modified version of EPROM (Erasable Programmable Read-Only Memory) which uses electrical signals to erase and program the contents rather than UV signals which was used previously in EPROM.
 - It is used as a chip in computers to store the digital data.
 - It is used as flash memories in its later version and also, to store BIOS of the computer.

Cache Memory:

- In between CPU and RAM
- Single or in Hierarchy
- Size of Cache increases



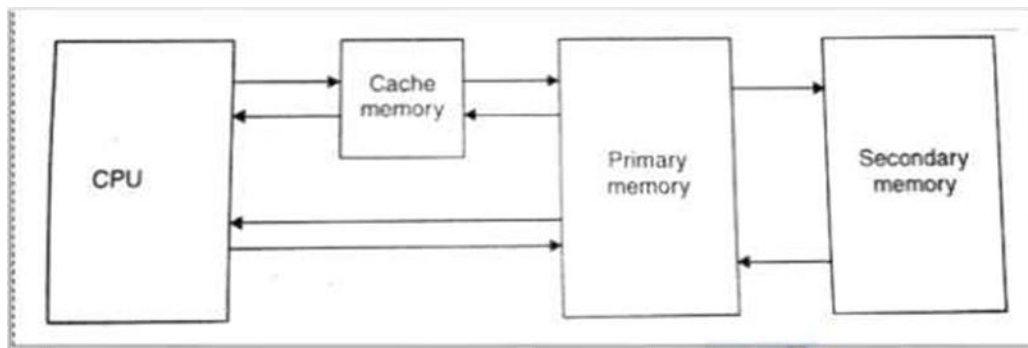
- Cache memory is a small, high-speed storage area in a computer. It stores copies of the data from frequently used main memory locations.
- There are various independent caches in a CPU, which store instructions and data.
- The most important use of cache memory is that it is used to reduce the average time to access data from the main memory.
- The concept of cache works because there exists locality of reference (the same items or nearby items are more likely to be accessed next) in processes.

Levels of Memory

- Level 1 or Register: It is a type of memory in which data is stored and accepted that are immediately stored in the CPU. The most

commonly used register is Accumulator, Program counter, Address Register, etc.

- Level 2 or Cache memory: It is the fastest memory that has faster access time where data is temporarily stored for faster access.
- Level 3 or Main Memory: It is the memory on which the computer works currently. It is small in size and once power is off data no longer stays in this memory.



Cache Performance

- When the processor needs to read or write a location in the main memory, it first checks for a corresponding entry in the cache.
- If the processor finds that the memory location is in the cache, a Cache Hit has occurred and data is read from the cache.
- If the processor does not find the memory location in the cache, a cache miss has occurred. For a cache miss, the cache allocates a new entry and copies in data from the main memory, then the request is fulfilled from the contents of the cache.
- The performance of cache memory is frequently measured in terms of a quantity called Hit ratio.

- $\text{Hit Ratio}(H) = \text{hit} / (\text{hit} + \text{miss}) = \text{no. of hits} / \text{total accesses}$
- $\text{Miss Ratio} = \text{miss} / (\text{hit} + \text{miss}) = \text{no. of miss} / \text{total accesses}$
 $= 1 - \text{hit ratio}(H)$

Locality issues for Cache

- **Locality of Reference (Memory Location)**
 - The reference to memory at any given interval of time tends to be confined within a few localized area of memory. This property is called locality of reference. This is possible because the program loops and subroutine calls are encountered frequently.
- **Spatial Locality (Memory Cluster)**
 - It refers to the tendency of execution to involve a number of memory locations that are clustered. It reflects tendency of a program to access data locations sequentially, such as when processing a table of data.
- **Temporal Locality (Frequency)**
 - It refers to the tendency for a processor to access memory locations that have been used frequently. For e.g. Iteration loops executes same set of instructions repeatedly.

Elements of Cache Design:

- **Cache Size**
- **Mapping Function**
- **Replacement Algorithm**
- **Write Policy**
- **Number of Caches**

Cache Mapping

Cache mapping refers to the method used to store data from main memory into the cache. It determines how data from memory is mapped to specific locations in the cache.

Cache mapping is the method by which the contents of main memory are brought into the cache and referenced by the CPU.

Key Terminologies in Cache Mapping

- **Main Memory Blocks:** The main memory is divided into equal-sized sections called blocks.
- **Cache Lines (or Cache Blocks):** The cache memory is also divided into equal partitions called cache lines.
- **Block Size:** The number of bytes or words stored in one block or line.
- **Tag Bits:** A small portion of the address used to identify which block of main memory is stored in a particular cache line.

There are three different types of mapping used for the purpose of cache memory which is as follows:

- **Direct Mapping**
- **Fully Associative Mapping**
- **Set-Associative Mapping**

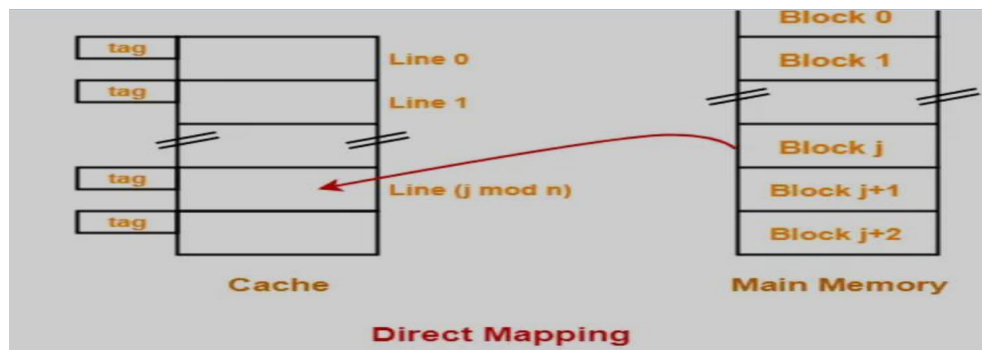
Direct mapping:

It is a procedure used to assign each memory block in the main memory to a particular line in the cache. If a line is already filled with a memory block and a new block needs to be loaded, then the old block is discarded from the cache.

A particular block of main memory can map only to a particular line of the cache.

The line number of cache to which a particular block can map is given by-

Consider cache memory is divided into 'n' number of lines. Then, block 'j' of main memory can map to line number $(j \bmod n)$ only of the cache.



Tag	Line Number	Block / Line Offset
Block Number		

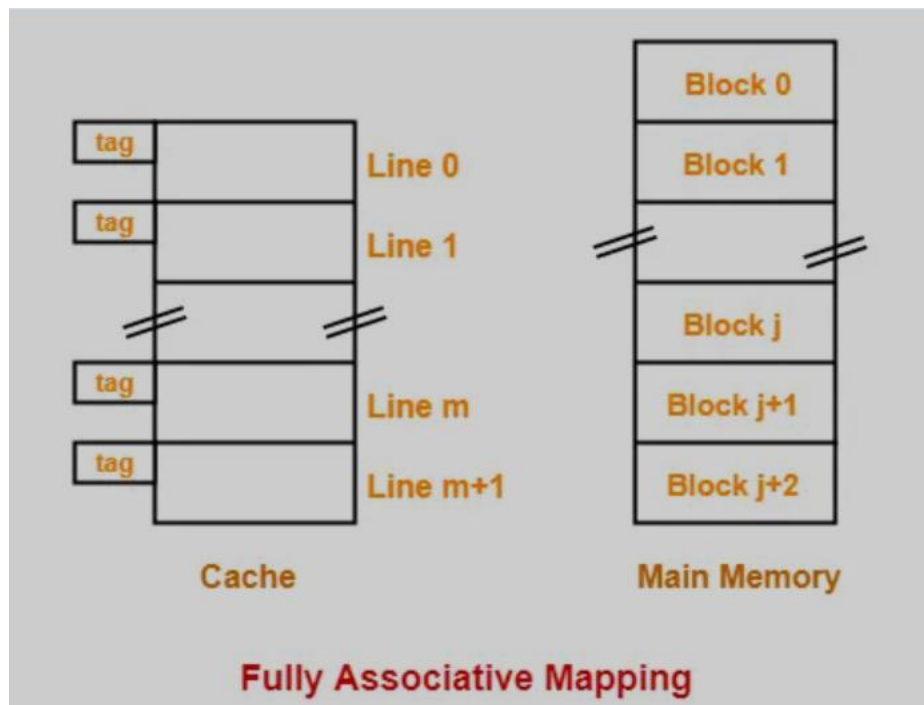
Division of Physical Address in Direct Mapping

Associative Mapping

Associative mapping is a more flexible cache mapping technique where a memory block can be stored in any cache line rather than being restricted to a specific one.

During access, the cache searches all lines in parallel for the tag that matches the requested memory address.

While this technique offers more flexibility and reduces the risk of cache thrashing, it requires more complex hardware to compare tags across all cache lines simultaneously.



Here,

All the lines of cache are freely available. Thus, any block of main memory can map to any line of the cache. Had all the cache lines been occupied, then one of the existing blocks will have to be replaced.



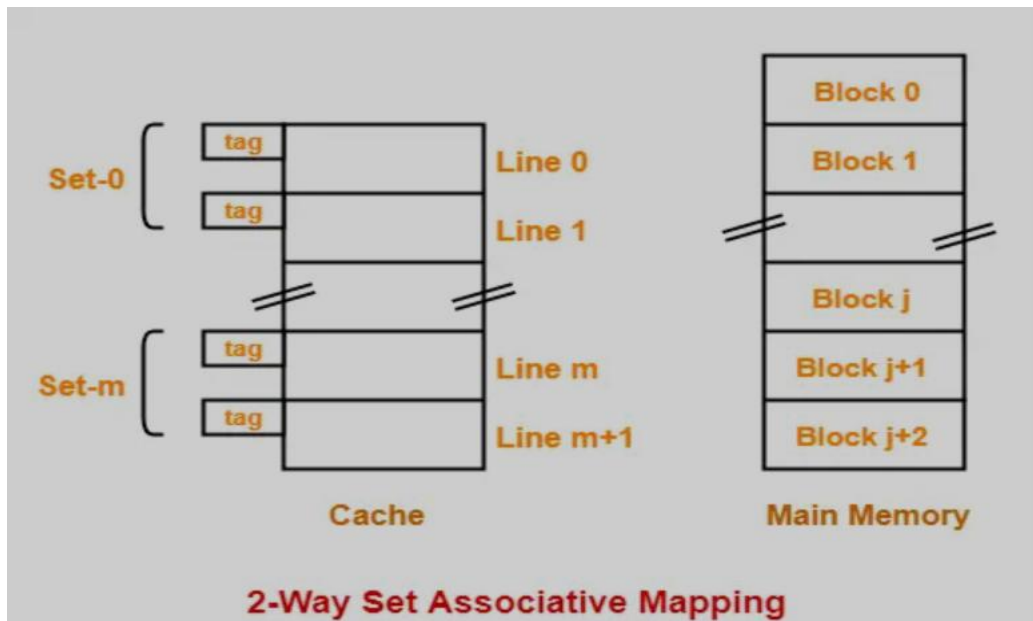
Set Associative Mapping:

In k-way set associative mapping, Cache lines are grouped into sets where each set contains k number of lines.

A particular block of main memory can map to only one particular set of the cache. However, within that set, the memory block can map any cache line that is freely available.

The set of the cache to which a particular block of the main memory can map is given by-

Cache set number= (Main Memory Block Address) Modulo (Number of sets in Cache)



Q. Consider a Cache with 256 blocks of 16 words each. The memory is addressed with 16 bits. How the address is divided or what is the tag size in

- Direct Mapping
- Associative Mapping
- Set Associative Mapping (2 way)

Solution:

Given:

- memory address is 16 bits (2^4 bits)
- Total Blocks 256 (2^8 blocks)
- Total Sets 128 (2^7)

For Direct Mapping:

Tag	Block	Words(offset)
4	8	4

For Associative Mapping:

Tag	Words(offset)
12	4

For Set Associative Mapping:

Tag	Set	Words(offset)
5	7	4

Q. A computer uses 32-bit addressing. The computer uses a 2 way associative cache with a capacity of 32KB. Each cache block contains 16 bytes. Calculate the number of bits in a tag, Set and offset fields of main memory.

Solution:

Given:

- Since there are 16 bytes in a cache block, the offset field must contain 4 bits ($2^4=16$).
- Each set contains 2 cache blocks(2-way Associative). So a set contains 32 bytes.
- There are 32KB bytes in the entire cache, so there are $32KB/32B= 1K$ sets. Thus the set field contains 10 bits ($2^{10}=1k$)

18	10	4
TAG	SET	OFFSET

Q. A computer system uses 16-bit memory addresses. It has a 2K-byte cache organized in a direct-mapped manner with 64 bytes per cache block. Assume that the size of each memory word is 1 byte.
(a) Calculate the number of bits in each of the Tag, Block, and Word fields of the memory address.

Solution:

- Block size = 64 bytes = 2^6 bytes = 2^6 words (since 1 word = 1 byte)
- Therefore, Number of bits in the Word field = 6
- Cache size = 2K-byte = 2^{11} bytes
- Number of cache blocks = Cache size / Block size = $2^{11}/2^6 = 2^5$
- Therefore, Number of bits in the Block field = 5
- Total number of address bits = 16
- Therefore, Number of bits in the Tag field = $16 - 6 - 5 = 5$
- For a given 16-bit address, the 5 most significant bits, represent the Tag, the next 5 bits represent the Block, and the 6 least significant bits represent the Word.

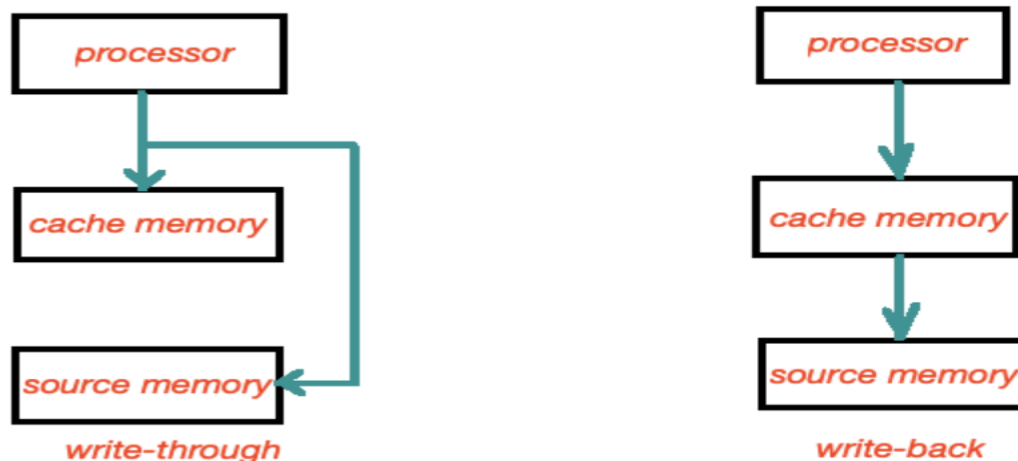
Replacing Algorithm in the cache

- For direct mapping where there is only one possible line for a block of memory, no replacement algorithm is needed.
- For associative and set associative mapping, however, an algorithm is needed.
- For maximum speed, this algorithm is implemented in the hardware. Four of the most common algorithms are:
 1. Least recently used this replaces the candidate line in cache memory that has been there the longest with no reference to it.
 2. First in first out this replaces the candidate line in the cache that has been there the longest.
 3. Least frequently used this replaces the candidate line in the cache that has had the fewest references.

4. Random replacement this algorithm randomly chooses a line to be replaced from among the candidate lines. Studies have shown that this yields only slightly inferior performance than other algorithms.

Write Policy

- When a line is to be replaced, must update the original copy of the line in main memory if any addressable unit in the line has been changed
- If a block has been altered in cache, it is necessary to write it back out to main memory before replacing it with another block
- Must not overwrite a cache block unless main memory is up to date
- I/O modules may be able to read/write directly to memory
- Multiple CPU's may be attached to the same bus, each with their own cache
- **Write-through**
 - All write operations are made to main memory as well as to cache, so main memory is always valid
 - Other CPU's monitor traffic to main memory to update their caches when needed
 - Anytime a word in cache is changed, it is also changed in main memory
 - Both copies always agree
 - Generates lots of memory writes to main memory
- **Write-back**
 - When an update occurs, an UPDATE bit associated with that slot is set, so when the block is replaced it is written back first
 - During a write, only change the contents of the cache
 - Update main memory only when the cache line is to be replaced



RAID (Redundant Arrays of Independent Disks)

RAID is a technique that makes use of a combination of multiple disks instead of using a single disk for increased performance, data redundancy, or both. If the data is spread across multiple disks without the RAID technique, the loss of a single disk can affect the entire data.

1. RAID-0 (Stripping)

- Blocks are “stripped” across disks.
- In the figure, blocks “0,1,2,3” form a stripe.
- Instead of placing just one block into a disk at a time, we can work with two (or more) blocks placed into a disk before moving on to the next one.
- It is easy to implement.
- It utilizes the storage capacity in a better way.

RAID 0

Disk 0	Disk 1	Disk 2	Disk 3
0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

2. RAID-1 (Mirroring)

- More than one copy of each block is stored in a separate disk. Thus, every block has two (or more) copies, lying on different disks.
- 1 disk failure can be handled for certain because blocks of that disk would have duplicates on some other disk
- RAID 0 was unable to tolerate any disk failure. But RAID 1 is capable of reliability.
- it can increase data security and speed.

RAID 1

Disk 0	Disk 1	Disk 2	Disk 3
0	0	1	1
2	2	3	3
4	4	5	5
6	6	7	7

RAID-3 (Byte-Level Striping with Dedicated Parity)

- It consists of byte-level striping with dedicated parity striping.
- At this level, we store parity information in a disc section and write to a dedicated parity drive.
- Whenever failure of the drive occurs, it helps in accessing the parity drive, through which we can reconstruct the data.

RAID 3

Disk 0	Disk 1	Disk 2	Disk 3
15	16	17	P(15, 16, 17)
18	19	20	P(18, 19, 20)
21	22	23	P(21, 22, 23)
24	25	26	P(24, 25, 26)

RAID-5 (Block-Level Striping with Distributed Parity)

This is a slight modification of the RAID-4 system where the only difference is that the parity rotates among the drives.

In the figure, we can notice how the parity bit “rotates”.

This was introduced to make the random write performance better.

RAID 5

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
5	6	7	P1	4
10	11	P2	8	9
15	P3	12	13	14
P4	16	17	18	19

DDR (Double Data Rate) RAM

DDR (Double Data Rate) DRAM is main memory using capacitors to store data and sense amplifiers + prefetch buffers to transfer data efficiently.

Transfers data on both rising and falling edges of the clock → double data rate vs old SDR RAM.

Old RAM = SDR (Single Data Rate)

Transfers 1 bit per clock cycle, using rising edge only.

DRAM stores data in tiny capacitors.

1 = charged

0 = discharged

Capacitors leak charge, so DRAM needs constant refreshing.

It Consist of three different component:

Sense Amplifier

A tiny circuit inside DRAM that reads the charge from capacitors.

Each capacitor is very small, so the voltage difference between 1 and 0 is tiny.

The sense amplifier amplifies this tiny voltage so the DRAM can reliably detect 1 or 0.

DRAM capacitor voltage is too weak to send directly to CPU.

Sense amplifier makes the signal strong enough to process.

Prefetch Buffer

Temporary storage inside DRAM to hold multiple bits read from capacitors before sending to CPU.

DDR uses prefetch to increase data per clock.

Capacitors are slow to access individually.

CPU wants continuous, fast stream of data.

Prefetch buffer stores multiple bits at once → can output data at DDR clock edges

Memory Controller

The brain of the memory system.

Which row/column to read/write

When to activate sense amplifiers

When to refresh capacitors

Timing for DDR data transfer

DIMM (Dual Inline Memory Module)

DIMM (Dual Inline Memory Module) is a physical memory module used to mount DRAM chips and connect them to the memory controller on the motherboard.

Dual Inline” means:

The electrical contacts (pins) on both sides of the module are independent.

Each side carries different signals.

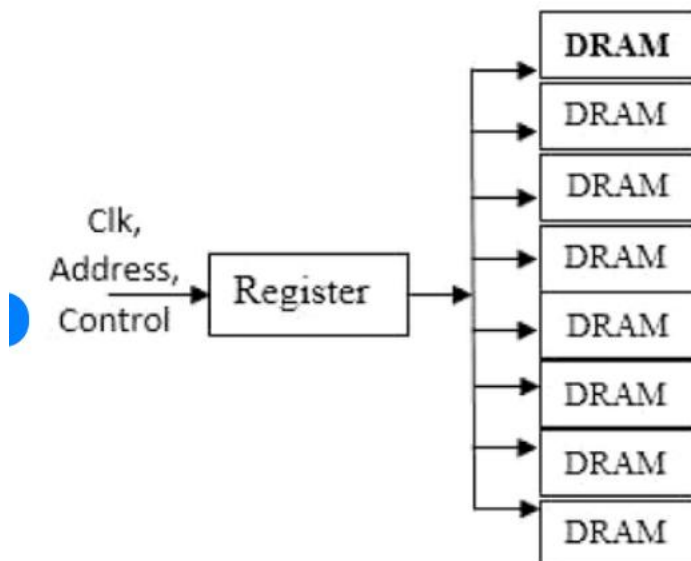
Earlier SIMM (Single Inline Memory Module):

Pins on both sides were electrically same → limited bandwidth.

DIMM is a module that contains one or several random access memory (RAM) chips on a small circuit board with pins that connect it to the computer motherboard. The DIMM stores each data bit in a separate memory cell.

DIMM (Dual In-Line Memory Module) is a type of RAM module used in computers to provide main memory.

- Provides high-speed data access to CPU
- Supports 64-bit data bus (directly matches modern processors)
- Allows larger memory capacity
- Better performance and reliability than older SIMM modules



Step 1: CPU Generates Memory Request

CPU needs data or instruction

Sends address, control signals (read/write) to the memory controller

Step 2: Memory Controller Accesses DIMM

Memory controller (inside CPU or chipset) decodes:

Row address

Column address

Activates the appropriate DIMM slot

Step 3: Row & Column Selection in DRAM

DIMM uses DRAM, which stores data in rows and columns:

RAS (Row Address Strobe)

Selects the row in memory

CAS (Column Address Strobe)

Selects the column in that row

This process identifies the exact memory cell.

Step 4: Data Transfer

For read operation:

Data is fetched from DRAM cell

Sent through 64-bit data bus to CPU

For write operation:

CPU sends data to DIMM

Data is stored in DRAM cells

Step 5: Refresh Operation

Since DIMM uses DRAM, data must be refreshed

Memory controller periodically refreshes rows to prevent data loss.

CPU issues a memory request

Memory controller selects a DIMM

Address and control signals go through DIMM pins

DRAM chips on DIMM:

- Read capacitor charge
- Use sense amplifiers
- Load data into prefetch buffers

Data transferred via **DDR clock** to CPU

SO-DIMM (Small Outline Dual Inline Memory Module)

SO-DIMM (Small Outline Dual Inline Memory Module) is a compact version of DIMM, designed to hold DRAM chips in devices where space and power consumption are limited.

It performs the same function as DIMM, but in a smaller physical size.

“Small Outline Dual Inline”

Small Outline → Smaller PCB size

Dual Inline → Pins on both sides are electrically independent

CPU issues a memory request

Memory controller selects an SO-DIMM slot

Address, control, and clock signals pass through SO-DIMM pins

DRAM chips:

Read capacitor charge

Sense amplifiers amplify data

Prefetch buffers prepare multiple bits

DDR transfers data on rising + falling clock edges

Data reaches CPU

Solid-State Drive (SSD)

A Solid-State Drive (SSD) is a non-volatile storage device that stores data without using any moving parts, unlike traditional Hard Disk Drives (HDDs), which have spinning disks and mechanical read/write heads. Because of this, SSDs are much faster, more durable and quieter than HDDs. They load files, programs and operating systems quickly, making computers more responsive.

Solid state drives (SSDs) use a combination of NAND flash memory technology and advanced controller algorithms. NAND flash memory is the primary storage component, divided into blocks and pages.

An SSD contains a controller chip that manages data storage, retrieval and optimization.

The controller's major duties are wear leveling, which evenly distributes write and erase cycles to extend the SSD's lifespan which consolidates empty blocks to maintain optimal performance.

Types of SSD

Single-Level Cell (SLC): Stores 1 bit per cell, offering the fastest speed, longest lifespan and highest reliability, but is expensive.

Multi-Level Cell (MLC): Stores 2 bits per cell, balancing speed, cost and durability. Commonly used in consumer and enterprise SSDs.

Triple-Level Cell (TLC) – Stores 3 bits per cell, making it more affordable but with slower speeds and a shorter lifespan than SLC and MLC.

Quad-Level Cell (QLC): Stores 4 bits per cell, offering the highest storage capacity at a lower cost but with reduced performance and endurance.

.Assignment Question:

- What is memory hierarchy? Describe the significance of cache and its Level. What is the Write policy in Cache.
- Differentiate between different types of cache mapping.
- Explain about the working of DRAM and SRAM with Proper Diagram.
- What is cache performance? Describe in brief the strategies used to write data to cache memory.
- What is cache? Do you agree that system works without cache? Give suitable reasons to support your answer.
- Describe the four most common replacement algorithms related to design issues of cache memory? Explain Using Example.
- What are the significance of cache memory? Write different types of mapping technique.
- Show the layout of the cache for CPU that can address $1M \times 16$ of memory; the cache hold $8K \times 16$ of data and has the following mapping strategies. Given the number of bits per location and total number of locations as well. If block size of 4bit.

Direct mapped

Two-way associative

Fully associative

- Describe how a 1KB cache is mapped directly with 1 MB main memory. Use suitable diagram to show the configuration and determine tag and index.
- Explain different design issues of cache memory.
- Write short Notes:
 - Memory and its types
 - DIMM
 - SSD, Optical Disk and Magnetic Tape
- Write about the working of DDR?
- Explain the Major Characteristics of Memory.